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Determination of free and protein-bound DA and NE and their metabolites and oxidation products by UPLC-MS/MS method V.1

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We use this protocol and it's working

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Abstract

Protocol for the determination of free and protein-bound DA and NE and their metabolites and oxidation products by UPLC-MS/MS method

Materials

External standards

DA, DOPA, NE, DOPAC, DOMA, DOPE, VMA → up to 3000 nM in 25 mM FA in water

3MT, AC, VMA, 5-SCDA, 5-SCD → up to 1000 nM in 25 mM FA in water

Internal standard (IS)

DA-4d → 500 nM in 25 mM FA in water

Preparation of the aminochrome (AC) external standard

1m

- 1 Mix  500 μL of [M] 1 millimolar (mM) dopamine (DA) with  500 μL of [M] 2 millimolar (mM) KIO₄ dissolved in [M] 100 micromolar (μM) aqueous ammonium acetate buffer  5.8 at RT with vigorous shaking for 1 min.

Note

Following oxidation, aminochromes are placed on ice to prevent further decomposition. Significant degradation of all aminochromes occurs at both RT and 4 °C within 24 h and -20°C at 24-48h (Ochs 2005; Lemos-Amado 2001).

Preparation of calibration curves

1h

- 2 Prepare a stock solution of the IS in [M] 25 millimolar (mM) FA and store it at -80 °C.

Prepare fresh solutions of each metabolite in [M] 25 millimolar (mM) FA and use them to make three mixtures: MIX1 (DA, L-DOPA, NE, 3MT, AC), MIX2 (DOMA, DOPE, DOPAC) and MIX3 (5SCD and 5SCDA).

- 3 Serially dilute mixtures with [M] 25 millimolar (mM) FA to obtain the concentration series used in calibration curves.

Note

Typically, final calibration levels cover a range of 1.72–3000 nM for DA, NE, and MIX2 and 0.39–1000 nM for L-DOPA, 3MT, AC and MIX3.

- 4 Homogenize control samples (i.e brain, intestines, heart, blood serum, cells...) in the appropriate volume of [M] 250 millimolar (mM) FA
- 5 Distribute the sample into  90 μL aliquots prior to the addition of  30 μL of the appropriate working mixture (MIX1, MIX2 or MIX3),  96 μL of



[M] 25 millimolar (mM) FA and  24 μL of [M] 8 micromolar (μM) IS .

6  20000 rcf, 4°C, 00:10:00

10m

7 Transfer supernatant to an Ostro protein precipitation and phospholipid removal plate (Waters, USA) to filter it.

Save the pellet for protein-bound determinations (see below)

8 Finally, inject  7 μL into the UPLC-MS/MS system.

Sample preparation

2h

9 Add  300 μL of [M] 250 millimolar (mM) FA to each brain, intestine, heart or cell pellet sample prior homogenization. Dilute blood serum samples 1:10

Note

Due to the poor stability of aminochrome, usually a maximum of 50 samples can be analyzed at a time

10 Take a  20 μL 20 μL sample for protein determination (diluted 1/5 in [M] 25 millimolar (mM) FA)

11 Take  240 μL for metabolite determination and add  26 μL of IS

Note

Important!!!: ensure the concentration of IS is exactly the same in both calibration curves and samples

12  20000 rcf, 4°C, 00:10:00

10m

**Note**

The supernatant is used to determine free neurotransmitters and metabolites (that is, those present in the deproteinated supernatant) while the pellet is used for protein-bound determinations (that is, those present in the acid-insoluble pellet and released by HCl hydrolysis)

13 Transfer supernatant to an Ostro protein precipitation and phospholipid removal plate (Waters, USA) to be filtered.

14 Inject  7 μL of filtered supernatant samples into the UPLC-MS/MS system

Reductive HCl hydrolysis of resulting pellets

18h

15

Safety information

Work in fume hood during all the procedure

After removal of the supernatant, wash the pellet (from both calibration curves and samples) with  1 mL of chloroform: methanol (1: 1, v/v) by vortex mixing

16  20000 rcf, 4°C, 00:10:00

10m

17 Transfer the resulting pellets to a sealed-capped tube with [M] 6 Mass Percent HCl containing [M] 5 % volume thioglycolic acid and [M] 1 Mass Percent phenol

Note

- **Calibration curves** --> add  280 μL of the mixture and  40 μL of the corresponding calibration curve working mixture

- **Samples** --> add  288 μL of the mixture and  32 μL of IS



- 18 Purge tubes with a stream of nitrogen, seal them and heat them at $110\text{ }^{\circ}\text{C}$ for $16:00:00$ 16h
- 19 Let tubes cool at $4\text{ }^{\circ}\text{C}$ for at least $00:30:00$ 30m
- 20 $20000\text{ rcf}, 4^{\circ}\text{C}, 00:10:00$ 10m
- 21 Treat the supernatant with with acid-washed alumina to extract catecholic compounds

Alumina extraction of catecholic compounds

1h

- 22 Transfer a $100\text{ }\mu\text{L}$ aliquot of each hydrolysate into a new Eppendorf tube containing 50 mg of acid-washed alumina and $200\text{ }\mu\text{L}$ of $1\text{ Mass Percent Na}_2\text{S}_2\text{O}_5$ - $1\text{ Mass Percent EDTA.2Na}$
- 23 Add $500\text{ }\mu\text{L}$ of $2.7\text{ Mass Percent Tris. HCl}$ - $2\text{ Mass Percent EDTA.2Na}$ $\text{pH } 9$ to the mixture
- 24 $1100\text{ rpm}, 22^{\circ}\text{C}, 00:05:00$ on a microtube mixer 5m
- 25 $20000\text{ rcf}, 00:10:00$ 10m
- 26 Remove the aqueous layer by aspiration and was alumina with 1 mL of Milli-Q water
- 27 $20000\text{ rcf}, 00:10:00$ 10m
- 28 Remove the aqueous layer by aspiration and was alumina with 1 mL of Milli-Q water

- 29  20000 rcf, 00:10:00 10m
- 30 Remove the aqueous layer by aspiration and was alumina with  1 mL of Milli-Q water
- 31  20000 rcf, 00:10:00 10m
- 32 Remove the aqueous layer by aspiration
- 33 Elute catechols from alumina with  100 μ L of [M] 0.4 Mass Percent HClO₄ by shaking for 2 min 2m
- 34 Collect all liquid into the injection plate without taking alumina
- Note

Alumina is discarded after extraction
- 35 Finally, inject  7 μ L into the UPLC-MS/MS system.

UPLC-MS/MS analysis

- 36
- A Waters Acquity™ UPLC system is coupled with a Xevo TQ-S triple quadrupole mass spectrometer with electrospray ionization interface (Waters). Instrument control, data acquisition, and analysis is performed using MassLynx V4.1 (Waters).
- Chromatographic separation of samples is performed on a Waters Acquity™ HSS T3 (1.8 μ m; 2.1×100mm) column coupled to an Acquity™ HSS T3 VanGuard (100Å, 1.8 μ m, 2.1 mm X 5 mm) pre-column (Waters). Column temperature is set at 45 °C and samples are maintained at 6 °C in the thermostatic autosampler.
- 37 The mobile phase consisted of solvent A (methanol 100%) and solvent B (25 mM FA in MQ water) at a flow of 0.4 mL/min with the following gradient profiles (depending on the

MIX):

MIX1 and MIX2:

0.5% B maintained for 0.5 min, 5% B at 0.9 min and maintained for 2.1min, 50% B at 2.8 min and maintained for 1.2 min, 0.5% B at 4.1 min followed by 0.2 min of equilibration. Total run time 4.3 min.

MIX3:

0.5% B maintained for 0.5 min, 8% B at 2.6 min, 50% B at 2.9 min and maintained for 0.6 min, 0.5% B at 3.7 min. Total run time 3.7 min

- 38 The mass spectrometer detector operates under the following parameters: source temperature 150 °C, desolvation temperature 450 °C, cone gas flow 50 L/hr, desolvation gas flow 1100 L/hr and collision gas flow 0.15 mL/min. Argon is used as the collision gas. The capillary voltage is set at 0.5 kV for MIX1 and MIX3, and at 2 kV for MIX2 detection. The electrospray ionization (ESI) source was operated in both positive and negative modes, depending on the analyte.